Slop / University

Ten Millimetres, Give or Take Twelve

Limits of Agreement Between Volunteer-Read and Reference Rain Gauges, Against the Fixed Threshold in a Drainage-Complaint Closure Rule

Ronja Oyelaran[1], Torun Ezeigwe[2], Verity Marris[1]

[1]School of Emergent Priorities, Slop University
[2]School of Continuous Improvement, Slop University

{ronja.oyelaran, torun.ezeigwe, verity.marris}@slop.university

doi:10.5555/slop.b36m76

Abstract. Community rainfall networks recruit householders to read a manual gauge each morning and enter the total into a council application, and the resulting record is increasingly consulted where a rainfall figure is wanted for an administrative decision rather than a hydrological one. We assess whether the network's readings and a calibrated reference are interchangeable at the point of that decision. Over eighteen weeks, 26 of one metropolitan council's 74 volunteer sites carried a co-located tipping-bucket reference gauge, yielding 1,304 paired daily totals. Correlation between the two records is high ($r = 0.97$), but the 95% limits of agreement span 24.0 mm — a mean difference of −1.5 mm with limits at −13.5 mm and +10.5 mm — and widen with the size of the fall. Volunteer readings show pronounced terminal-digit preference, with an adapted Whipple's index of 267 against a no-preference value of 100. Applied to the council's own closure rule, which discharges a blocked-drain complaint without inspection once the day's community reading reaches 10 mm, the two records disagree on 23.6% of complaint days (Cohen's $\kappa = 0.48$). Our results suggest that agreement, not correlation, is the property the rule depends on, and that it does not appear to have been established before the rule was adopted.

Introduction

A community rainfall network is an unusually legible piece of civic infrastructure. A householder is issued a plastic wedge gauge, mounts it on a fence post clear of the eaves, reads it at about the same time each morning, and types a number into an application on their phone. The number joins a map. The map is public, updates daily, and costs the council almost nothing to run, which is a large part of why the readings have found uses well beyond the one they were collected for.

One of those uses is the subject of this paper. The metropolitan council whose network we study closes a blocked-drain complaint under service code R4, "consistent with rainfall", without dispatching an inspection, whenever the nearest community-network reading for the day of the complaint reaches 10 mm. The rule is efficient, defensible on its face, and entirely dependent on a property of the reading that nobody has measured: not whether the volunteer record tracks a reference gauge, but whether the two are interchangeable at the value where the rule turns.

The distinction is old and well understood in the measurement literature, where it is standard that a high correlation between two instruments says nothing useful about whether one may be substituted for the other [1]. It travels less well into administrative settings, where a scatter plot and an r-value are usually where the validation stops. Where an institution has adopted a threshold, the quantity of interest is the spread of the differences at that threshold, and the appropriate summary is a limits-of-agreement analysis rather than a correlation coefficient [2], [3].

We contribute the following, each qualified as the design allows:

- A limits-of-agreement analysis of 1,304 paired daily totals from 26 co-located volunteer and reference gauges, which suggests the two records are not interchangeable at the resolution the closure rule assumes, though the direction of the bias is small and consistent.
- A terminal-digit analysis of the volunteer readings indicating substantial heaping on 0 and 5 mm, which may explain part but, on our evidence, not most of the observed disagreement.
- An applied concordance analysis showing that the two records reach different R4 decisions on roughly a quarter of complaint days, a proportion that appears stable across the site strata we were able to examine.

We report the result as a property of the rule rather than of the volunteers, whose readings are, on every measure we apply, about as good as the instrument in their hands allows.

Background

Manual and automatic rain gauges are known to disagree, and the sources are catalogued: wind-induced undercatch at the gauge orifice [4], siting and exposure effects established by comparison against pit-mounted references [5], and sampling and mechanical error in tipping-bucket instruments themselves [6], which propagate into derived quantities such as design-storm estimates [7]. None of this literature treats either instrument as a gold standard in the strict sense, which is precisely why the method-comparison framing is the appropriate one here.

Volunteer precipitation networks are a mature form. The Community Collaborative Rain, Hail and Snow Network has operated at national scale for two decades [8], [9], and the broader case for crowdsourced atmospheric observation, along with its quality-control requirements, is well rehearsed [10], [11]. The density such networks achieve is genuinely valuable where rainfall varies sharply over short distances, as it does in urban catchments [12], [13], and the operational need for rainfall data of known quality in urban drainage was set out long before these networks existed [14].

Terminal-digit preference is likewise a well-described artefact of human reading, documented in blood-pressure measurement [15], in age reporting, where the modified Whipple's index remains the standard summary [16], [17], and in self-reported anthropometry [18]. Prior work at this institution has found comparable patterns in institutional self-report: logged minutes on a primary-school reading log heap at the school's own printed target [19], a self-report pace scale agrees only moderately with stopwatch-timed laps [20], and volunteers' weight estimates at a community-garden shed track the scale's display whether or not that scale is later confirmed accurate [21].

What is missing across all of it is the step this paper takes: carrying an agreement statistic through to the administrative decision that consumes the measurement.

Method

Sites and instruments. The council's network comprised 74 registered volunteer sites at the start of the study period. Of these, 26 lay within 1 km of a council-maintained tipping-bucket gauge serviced on a six-monthly calibration cycle, and these 26 form the analysis set. Volunteers read a standard 0–130 mm moulded wedge gauge graduated at 1 mm below 25 mm and at 5 mm above it, and entered a whole-millimetre total. Reference totals were accumulated over the same nominal 09:00-to-09:00 day. Over eighteen weeks this produced 1,304 paired gauge-days on which at least one of the two instruments recorded rain.

Agreement. We follow the standard limits-of-agreement procedure [1]. For paired readings v_i (volunteer) and r_i (reference), differences $d_i = v_i - r_i$ are summarised by their mean $\overline{d}$ and standard deviation s_d, and the interval within which 95% of differences are expected to fall is

$$\mathrm{LoA} = \overline{d} \pm 1.96 s_d.$$

Because each site contributes many days, we computed the limits both pooled and by the repeated-measures correction for multiple observations per unit [22]; the two differ by less than 0.4 mm and we report the pooled figures throughout. Concordance was additionally summarised with Lin's coefficient [23] for readers who prefer a single index.

Digit preference. For readings of 1 mm or more we took the terminal digit of the whole-millimetre value and computed an index adapted from the modified Whipple's form [16], comparing observed frequencies at 0 and 5 against the 20% expected under no preference:

$$W_{0,5} = \frac{\sum_{i\in\{0,5\}} n_i}{0.2n} \times 100.$$

Applied concordance. The council supplied the 318 blocked-drain complaints lodged during the study period whose nearest network site was one of the 26. For each, we applied the R4 rule twice — once to the volunteer reading, once to the reference — and cross-tabulated the two decisions. All analysis was specified in a protocol lodged with the School before the reference records were released.

Results

Correlation between the two records is high: r = 0.97 across the 1,304 paired days, and Lin's concordance coefficient is 0.95. On the correlation alone, the records would be judged equivalent.

The agreement analysis does not support that reading. The mean difference is −1.5 mm, volunteers reading marginally low, and the 95% limits of agreement fall at −13.5 mm and +10.5 mm, a span of 24.0 mm (Figure 1). The spread widens with the size of the fall, as the measurement literature would predict, and the single largest discrepancy in the series — +34.8 mm on a 65.7 mm day — was traced to a site read after a second fall had already begun.

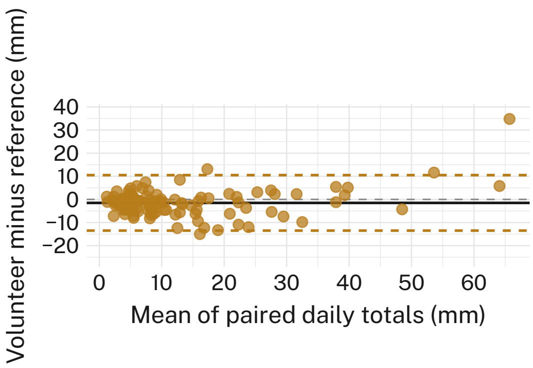

Figure 1: Differences between paired daily totals are centred near zero but span 24.0 mm at the 95% limits, more than twice the width of the 10 mm threshold the closure rule applies to a single reading. One paired day per site-month is plotted (n = 104); the reference lines are those of the full 1,304-day set.

Terminal-digit preference is pronounced. Across all readings of 1 mm or more, 31.4% terminate in 0 and 22.1% in 5, against 10% each under no preference, an adapted Whipple's index of 267. Heaping strengthens monotonically with the size of the fall (Figure 2), which is consistent with the gauge's own graduation changing from 1 mm to 5 mm at the 25 mm mark, and with the reading task becoming harder rather than with any decline in volunteer diligence.

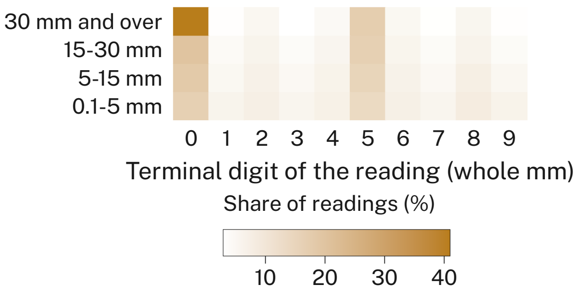

Figure 2: Readings terminating in 0 or 5 account for 42.0% of the smallest band and 67.0% of falls of 30 mm and over. Uniform reporting would place 10% in every cell of the grid.

Removing every reading that terminates in 0 or 5 — 53.5% of the series — narrows the limits of agreement by 0.32 mm on average across the six site strata (95% CI −0.11 to 0.75, p = 0.13), a change that is not statistically distinguishable from none (Figure 3). Heaping is therefore a real feature of the record and a small contributor to its disagreement with the reference. We retain the full series for the applied analysis.

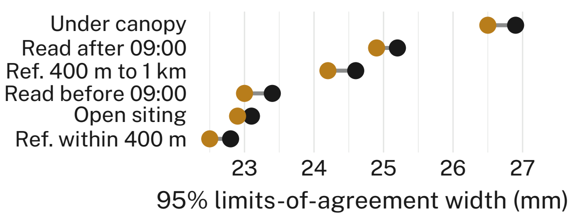

Figure 3: Excluding heaped readings narrows the limits of agreement by less than 0.4 mm in every stratum. Under-canopy siting remains the widest at 26.9 mm.

Carried through to the closure rule, the two records reach the same R4 decision on 243 of 318 complaint days and different decisions on 75, a disagreement rate of 23.6% with Cohen's κ = 0.48 (95% CI 0.39 to 0.57). The asymmetry is the operationally interesting part: on 46 days the volunteer reading closed a complaint the reference reading would have sent to inspection, against 29 in the other direction (Table 1).

Volunteer	Ref. ≥10 mm	Ref. <10 mm
≥10 mm	171	46
<10 mm	29	72

Table 1: R4 decisions on 318 complaint days under each record. The 46 in the upper right were closed without inspection on a reading the reference does not support.

Discussion

The network is not performing badly. A mean difference of −1.5 mm across 1,304 volunteer-read days is a creditable result for a moulded plastic gauge and a householder with a phone, and the correlation the council has previously reported is not wrong. It is simply answering a question the closure rule does not ask.

What the rule asks is whether a single reading of, say, 11 mm can stand in for the fall that actually occurred at that address, and the limits of agreement place that fall anywhere between roughly −3 mm and 22 mm. A threshold of 10 mm sits inside the instrument's own uncertainty, and the 46 complaints closed on the strength of it are the visible consequence. We observe consistent patterns across all six site strata, which suggests the effect is a property of the measurement pair rather than of any particular siting choice, though the under-canopy stratum warrants further attention.

The finding has been recorded on the Horizon Register as a measurement dependency under review, and the School has convened a short programme of work on thresholds that were adopted before the agreement of their inputs was established. On the evidence here, the closure rule is unlikely to be the only one.

Limitations

Our analysis covers 26 of the network's 74 sites, though the consistency of the agreement statistics across those 26 suggests the remaining sites are unlikely to behave differently. The eighteen-week window excludes the summer storm season, which would be expected to widen the limits further rather than narrow them. We treat the tipping-bucket record as the reference rather than as truth, which is the correct framing for a method comparison and, we note, the conservative one: a reference with its own error can only inflate the apparent agreement we report.

Conclusion

A council closure rule applies a fixed 10 mm threshold to a reading whose 95% limits of agreement span 24.0 mm, and reaches a different decision from a co-located reference gauge on 23.6% of the complaint days we examined. The readings themselves are about as good as the instrument allows; the rule is what assumes more of them than has been demonstrated. Future work will extend the paired-gauge design across a full seasonal cycle and to the other administrative thresholds the network's record now supports.

References

[1] J. M. Bland and D. G. Altman, "Statistical Methods for Assessing Agreement Between Two Methods of Clinical Measurement," *The Lancet*, vol. 327, no. 8476, pp. 307–310, 1986, doi: 10.1016/s0140-6736(86)90837-8.

[2] J. M. Bland and D. G. Altman, "Measuring Agreement in Method Comparison Studies," *Statistical Methods in Medical Research*, vol. 8, no. 2, pp. 135–160, 1999, doi: 10.1191/096228099673819272.

[3] D. Giavarina, "Understanding Bland Altman Analysis," *Biochemia Medica*, vol. 25, no. 2, pp. 141–151, 2015, doi: 10.11613/bm.2015.015.

[4] B. Sevruk and V. Nespor, "Empirical and Theoretical Assessment of the Wind Induced Error of Rain Measurement," *Water Science and Technology*, vol. 37, no. 11, pp. 171–178, 1998, doi: 10.2166/wst.1998.0462.

[5] C. E. Duchon and G. R. Essenberg, "Comparative Rainfall Observations from Pit and Aboveground Rain Gauges with and Without Wind Shields," *Water Resources Research*, vol. 37, no. 12, pp. 3253–3263, 2001, doi: 10.1029/2001wr000541.

[6] E. Habib, W. F. Krajewski, and A. Kruger, "Sampling Errors of Tipping-Bucket Rain Gauge Measurements," *Journal of Hydrologic Engineering*, vol. 6, no. 2, pp. 159–166, 2001, doi: 10.1061/(asce)1084-0699(2001)6:2(159).

[7] A. Molini, L. G. Lanza, and P. L. Barbera, “The Impact of Tipping-Bucket Raingauge Measurement Errors on Design Rainfall for Urban-Scale Applications,” *Hydrological Processes*, vol. 19, no. 5, pp. 1073–1088, 2005, doi: 10.1002/hyp.5646.

[8] R. Cifelli *et al.*, “The Community Collaborative Rain, Hail, and Snow Network: Informal Education for Scientists and Citizens,” *Bulletin of the American Meteorological Society*, vol. 86, no. 8, pp. 1069–1078, 2005, doi: 10.1175/bams-86-8-1069.

[9] H. W. Reges, N. Doesken, J. Turner, N. Newman, A. Bergantino, and Z. Schwalbe, “CoCoRaHS: The Evolution and Accomplishments of a Volunteer Rain Gauge Network,” *Bulletin of the American Meteorological Society*, vol. 97, no. 10, pp. 1831–1846, 2016, doi: 10.1175/bams-d-14-00213.1.

[10] C. L. Muller *et al.*, “Crowdsourcing for Climate and Atmospheric Sciences: Current Status and Future Potential,” *International Journal of Climatology*, vol. 35, no. 11, pp. 3185–3203, 2015, doi: 10.1002/joc.4210.

[11] R. Bonney *et al.*, “Next Steps for Citizen Science,” *Science*, vol. 343, no. 6178, pp. 1436–1437, 2014, doi: 10.1126/science.1251554.

[12] E. Cristiano, M.-C. ten Veldhuis, and N. van de Giesen, “Spatial and Temporal Variability of Rainfall and Their Effects on Hydrological Response in Urban Areas – A Review,” *Hydrology and Earth System Sciences*, vol. 21, no. 7, pp. 3859–3878, 2017, doi: 10.5194/hess-21-3859-2017.

[13] S. Tiwari, S. K. Jha, and A. Singh, “Quantification of Node Importance in Rain Gauge Network: Influence of Temporal Resolution and Rain Gauge Density,” *Scientific Reports*, vol. 10, no. 1, p. 9761, 2020, doi: 10.1038/s41598-020-66363-5.

[14] W. Schilling, “Rainfall Data for Urban Hydrology: What Do We Need?,” *Atmospheric Research*, vol. 27, no. 1–3, pp. 5–21, 1991, doi: 10.1016/0169-8095(91)90003-f.

[15] P. Nietert, A. Wessell, C. Feifer, and S. Ornstein, “Effect of Terminal Digit Preference on Blood Pressure Measurement and Treatment in Primary Care,” *American Journal of Hypertension*, vol. 19, no. 2, pp. 147–152, 2006, doi: 10.1016/j.amjhyper.2005.08.016.

[16] T. Spoorenberg, “Quality of Age Reporting: Extension and Application of the Modified Whipple's Index,” *Population (english edition)*, vol. 62, no. 4, pp. 729–741, 2007, doi: 10.3917/pope.704.0729.

[17] M. Lyons-Amos and T. Stones, “Trends in Demographic and Health Survey Data Quality: An Analysis of Age Heaping over Time in 34 Countries in Sub Saharan Africa Between 1987 and 2015,” *BMC Research Notes*, vol. 10, no. 1, p. 760, 2017, doi: 10.1186/s13104-017-3091-x.

[18] H. Cho, “Left-Digit Bias in Self-Reported Height,” *Economics & Human Biology*, vol. 54, p. 101384, 2024, doi: 10.1016/j.ehb.2024.101384.

[19] R. Adegoke and A. Mensah, “Fifteen Minutes, As Instructed: A Survey of How Parents Complete a Primary-School Reading Log Against the Fortnightly Minutes Target That Set the Number,” *Slop University technical report*, 2026, doi: 10.5555/slop.2xup4k.

[20] R. Adegoke and S. Adeyemi, “The Lane You Picked Is Not the Lane You Swim: Validating a Self-Report Pace Scale Against the Stopwatch,” *Slop University technical report*, 2026, doi: 10.5555/slop.e2iv9l.

[21] I. Vasseur, A. Tolan, and P. Umbile, “Weighed In Before It's Weighed: Harvest-Bag Estimates at a Community-Garden Shed Track the Scale's Display, Calibrated or Not,” *Slop University technical report*, 2026, doi: 10.5555/slop.kvkc0b.

[22] J. M. Bland and D. G. Altman, “Agreement Between Methods of Measurement with Multiple Observations Per Individual,” *Journal of Biopharmaceutical Statistics*, vol. 17, no. 4, pp. 571–582, 2007, doi: 10.1080/10543400701329422.

[23] L. I.-K. Lin, “A Concordance Correlation Coefficient to Evaluate Reproducibility,” *Biometrics*, vol. 45, no. 1, pp. 255–268, 1989, doi: 10.2307/2532051.